机器学习

第 5 章 神经网络

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计算机与人工智能学院

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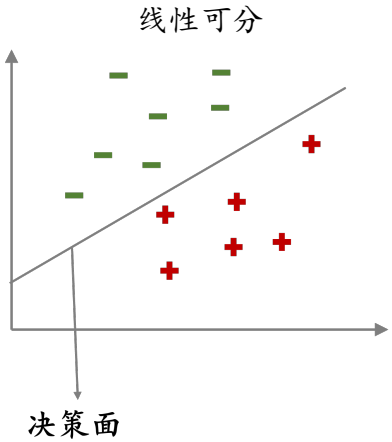
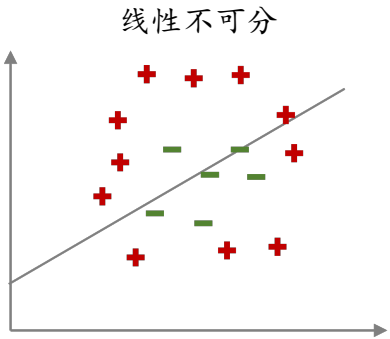
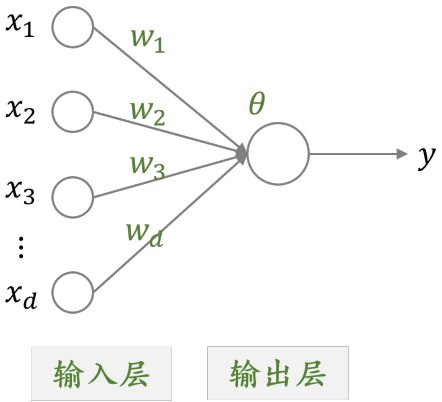
1. 感知机模型：线性可分与线性不可分



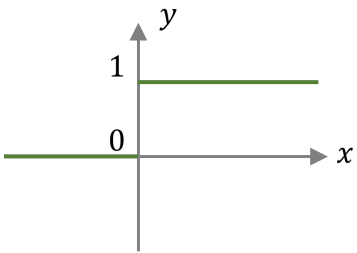


1. 感知机模型

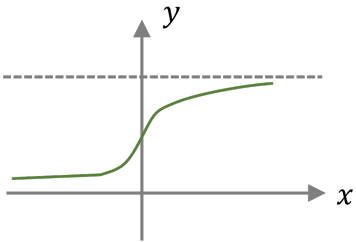


$$
\begin{aligned}
y &= f(w_1x_1 + w_2x_2 + \cdots + w_dx_d - \theta) \\
&= f\left(\sum_{i=1}^{d} w_ix_i - \theta\right) \\
&= f\left(\boldsymbol{w}^{\mathrm{T}}\boldsymbol{x} - \theta\right)
\end{aligned}
$$

1. 感知机模型：激活函数



sgn阶跃函数

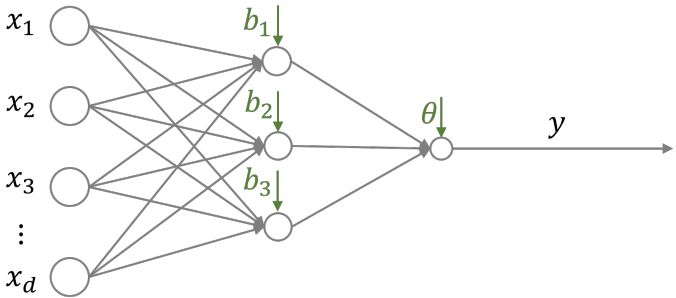


sigmoid函数

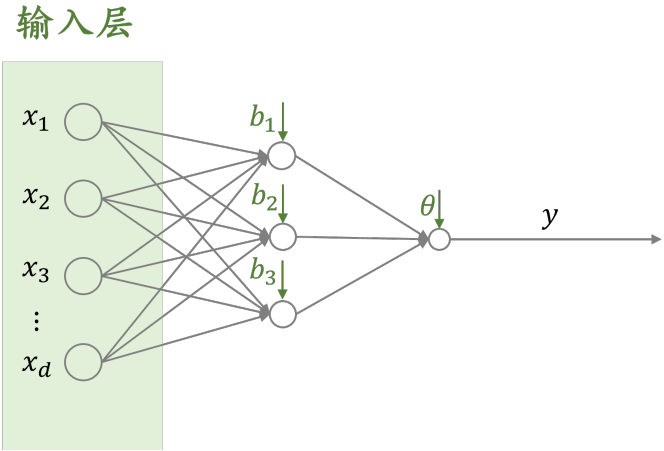
$$y(\boldsymbol{x}) = \mathrm{sgn}\left(\boldsymbol{w}^{\mathrm{T}}\boldsymbol{x} - \theta\right)$$

$$y(\boldsymbol{x}) = \mathrm{sigmoid}\left(\boldsymbol{w}^{\mathrm{T}}\boldsymbol{x} - \theta\right) = \frac{1}{1 + e^{(-\boldsymbol{w}^{\mathrm{T}}\boldsymbol{x} + \theta)}}$$

2. 神经网络：基本结构

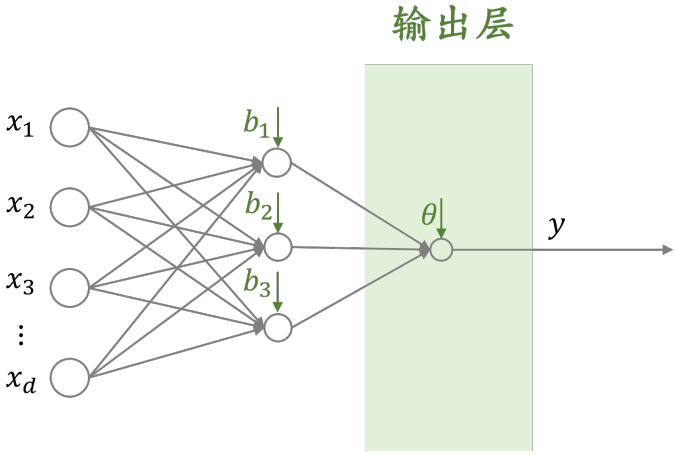


2. 神经网络：基本结构



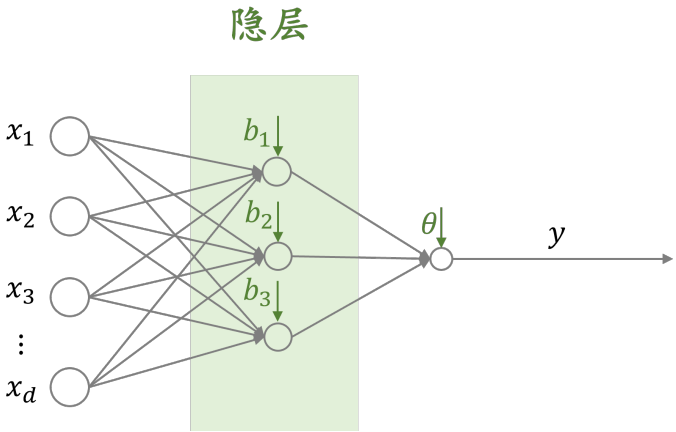
- 输入层没有神经元
- 节点数与特征数目相同

2. 神经网络：基本结构



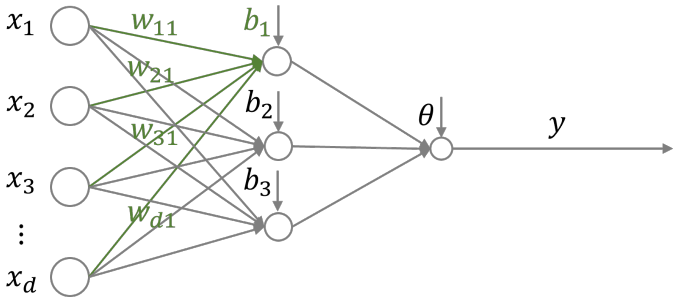
- 输出层有神经元
- 回归问题 1 个节点，k 分类问题 k 个节点

2. 神经网络：基本结构

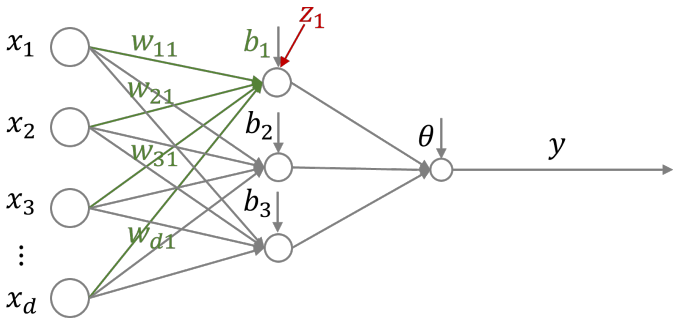


- 隐含层有神经元
- 隐含层层数，每层节点数可以看作超参数

2. 神经网络：前向传播



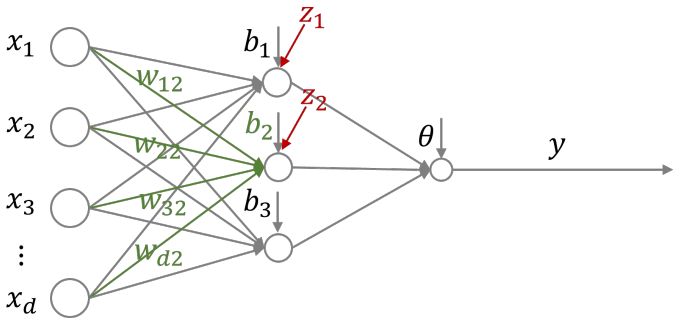
2. 神经网络：前向传播



$$z_1 = w_{11}x_1 + w_{21}x_2 + \cdots + w_{d1}x_d - b_1$$

$$z_1 = \boldsymbol{w}_1^{\mathrm{T}}\boldsymbol{x} - b_1, \quad \boldsymbol{w}_1 = [w_{11}, w_{21}, \cdots, w_{d1}]^{\mathrm{T}}$$

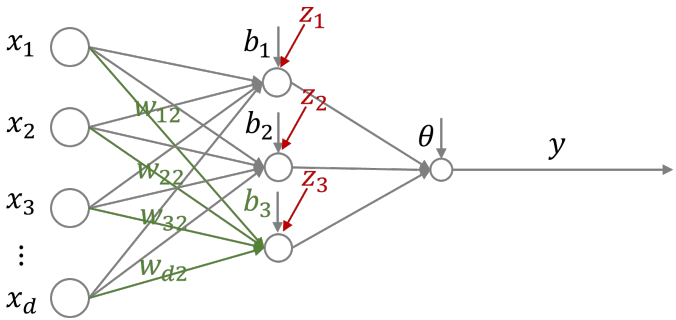
2. 神经网络：前向传播



$$z_1 = \boldsymbol{w}_1^{\mathrm{T}}\boldsymbol{x} - b_1, \quad \boldsymbol{w}_1 = [w_{11}, w_{21}, \cdots, w_{d1}]^{\mathrm{T}}$$

$$z_2 = \boldsymbol{w}_2^{\mathrm{T}}\boldsymbol{x} - b_2, \quad \boldsymbol{w}_2 = [w_{12}, w_{22}, \cdots, w_{d2}]^{\mathrm{T}}$$

2. 神经网络：前向传播

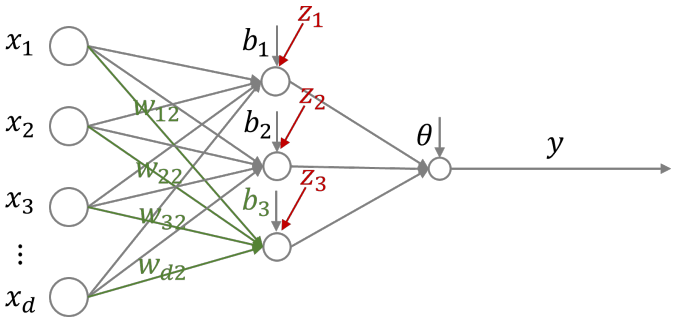


$$z_1 = \boldsymbol{w}_1^{\mathrm{T}}\boldsymbol{x} - b_1, \quad \boldsymbol{w}_1 = [w_{11}, w_{21}, \cdots, w_{d1}]^{\mathrm{T}}$$

$$z_2 = \boldsymbol{w}_2^{\mathrm{T}}\boldsymbol{x} - b_2, \quad \boldsymbol{w}_2 = [w_{12}, w_{22}, \cdots, w_{d2}]^{\mathrm{T}}$$

$$z_3 = \boldsymbol{w}_3^{\mathrm{T}}\boldsymbol{x} - b_3, \quad \boldsymbol{w}_3 = [w_{13}, w_{23}, \cdots, w_{d3}]^{\mathrm{T}}$$

2. 神经网络：前向传播



$$z_1 = \boldsymbol{w}_1^{\mathrm{T}}\boldsymbol{x} - b_1$$

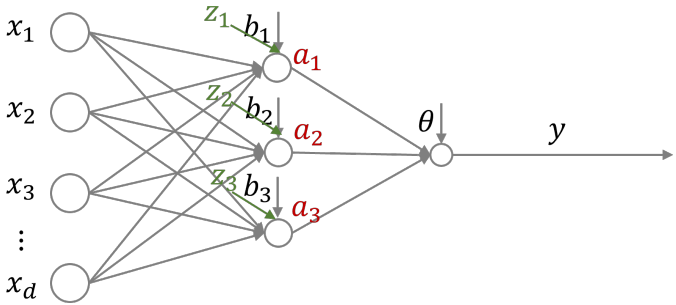
$$z_2 = \boldsymbol{w}_2^{\mathrm{T}}\boldsymbol{x} - b_2$$

$$z_3 = \boldsymbol{w}_3^{\mathrm{T}}\boldsymbol{x} - b_3$$

$$\boldsymbol{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_d \end{bmatrix} \quad \boldsymbol{W} = \begin{bmatrix} \boldsymbol{w}_1 & \boldsymbol{w}_2 & \boldsymbol{w}_3 \end{bmatrix} = \begin{bmatrix} w_{11} & w_{21} & w_{31} \\ w_{21} & w_{22} & w_{32} \\ \vdots & \vdots & \vdots \\ w_{d1} & w_{d2} & w_{d3} \end{bmatrix} \quad \boldsymbol{b} = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix} \quad \boldsymbol{z} = \begin{bmatrix} z_1 \\ z_2 \\ z_3 \end{bmatrix}$$

$$\boldsymbol{z} = \boldsymbol{W}^{\mathrm{T}}\boldsymbol{x} - \boldsymbol{b}$$

2. 神经网络：前向传播



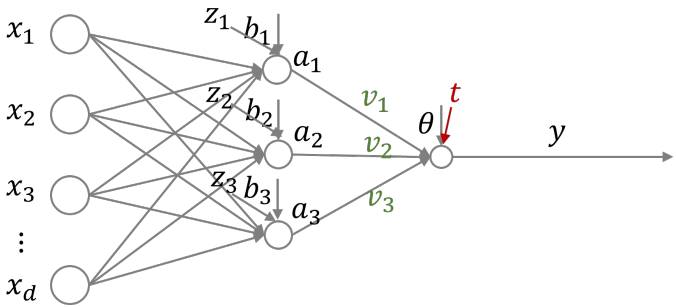
$$a_1 = \sigma(z_1)$$

$$a_2 = \sigma(z_2)$$

$$a_3 = \sigma(z_3)$$

$$\boldsymbol{a} = \boldsymbol{\sigma}(\boldsymbol{z})$$

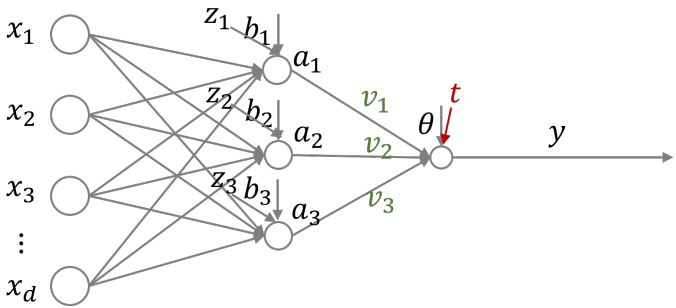
2. 神经网络：前向传播



$$t = \boldsymbol{v}^{\mathrm{T}}\boldsymbol{a} - \theta$$

$$y = \sigma(t)$$

2. 神经网络：前向传播



$$\boldsymbol{z} = \boldsymbol{W}^{\mathrm{T}}\boldsymbol{x} - \boldsymbol{b}$$

$$\boldsymbol{a} = \boldsymbol{\sigma}(\boldsymbol{z})$$

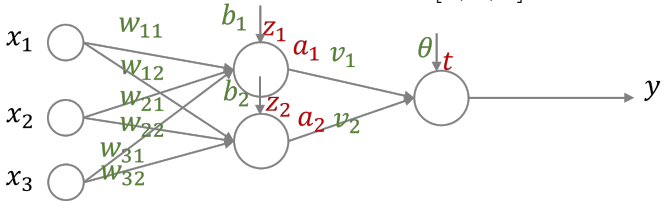
$$t = \boldsymbol{v}^{\mathrm{T}}\boldsymbol{a} - \theta$$

$$y = \sigma(t)$$

$$y = \sigma\left(\boldsymbol{v}^{\mathrm{T}}\boldsymbol{\sigma}\left(\boldsymbol{W}^{\mathrm{T}}\boldsymbol{x} - \boldsymbol{b}\right) - \theta\right)$$

2. 神经网络：作业 5

有如图所示的神经网络结构，网络所有节点激活函数均为 sigmoid 函数，其中 $w_{11}=2$，$w_{21}=1$，$w_{31}=4$，$w_{12}=-2$，$w_{22}=-3$，$w_{32}=1$，$b_1=14$，$b_2=-4$，$v_1=1$，$v_2=2$，$\theta=2$。试计算神经网络对输入 $[1,2,3]^{\mathrm{T}}$ 的输出值。



$$z_1 = w_{11}x_1 + w_{21}x_2 + w_{31}x_3 - b_1 = 2\times 1 + 1\times 2 + 4\times 3 - 14 = 2$$

$$z_2 = w_{12}x_1 + w_{22}x_2 + w_{32}x_3 - b_2 = -2\times 1 - 3\times 2 + 1\times 3 - (-4) = -1$$

$$a_1 = \sigma(z_1) = 0.8808, a_2 = \sigma(z_2) = 0.2689$$

$$t = v_1a_1 + v_2a_2 - \theta = 1\times 0.8808 + 2\times 0.2689 - 2 = -0.5814$$

$$y = \sigma(t) = 0.3586$$